ASESII 24A02 Project-Based Quiz 1

Ridge, Lasso, and Regularization for Neural Features

Group XX: [names and student IDs]

2026-07-15

Contents

Authorship and contribution statement

We confirm that every group member can explain the submitted analysis. AI use is reported in `Group04_AI_Log.csv`, and the group checked every AI-assisted item used in this report.

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Abstract

Maximum 150 words: prediction problem, methods, main evidence, and recommendation.

Reproducible setup

```r
required_packages <- c("tidyverse", "glmnet", "broom", "knitr")
missing_packages <- required_packages[
  !vapply(required_packages, requireNamespace, logical(1), quietly = TRUE)
]
if (length(missing_packages)) {
  stop("Install the required packages before rendering: ",
       paste(missing_packages, collapse = ", "))
```

```r
}

library(tidyverse)
library(glmnet)
library(broom)
library(knitr)

group_number <- as.integer(params$group_number)
if (length(group_number) != 1L || is.na(group_number) || group_number < 1L) {
  stop("Set params$group_number to your assigned positive group number.")
}

seeds <- list(
  split = 240200L + group_number,
  cv = 240300L + group_number,
  features = 240400L + group_number
)
```

Shared helper functions

```r
find_exam_data <- function(filename) {
  input <- knitr::current_input(dir = TRUE)
  input_dir <- if (length(input) && nzchar(input)) dirname(input) else getwd()
  candidates <- c(
    file.path(input_dir, "data", filename),
    file.path(input_dir, "..", "data", filename),
    file.path(getwd(), "data", filename)
  )
  existing <- candidates[file.exists(candidates)]
  if (!length(existing)) stop("Cannot locate ", filename, " by a relative path.")
  hits <- unique(normalizePath(existing, winslash = "/", mustWork = TRUE))
  if (length(hits) > 1L && length(unique(unname(tools::md5sum(hits)))) > 1L) {
    stop("Multiple nonidentical copies of ", filename, " were found.")
  }
  hits[[1L]]
}

split_rows <- function(n, train_prop = 0.80, seed) {
  stopifnot(n >= 10L, train_prop > 0, train_prop < 1)
  set.seed(seed)
  train <- sort(sample.int(n, floor(train_prop * n), replace = FALSE))
  list(train = train, test = setdiff(seq_len(n), train))
}

fit_scaler <- function(x, tol = 1e-12) {
```

```r
  x <- as.matrix(x)
  center <- colMeans(x)
  centered <- sweep(x, 2L, center, "-")
  scale <- sqrt(colMeans(centered^2))
  keep <- is.finite(scale) & scale > tol
  if (!any(keep)) stop("No nonconstant training predictor remains.")
  list(
    center = center[keep],
    scale = scale[keep],
    keep = keep,
    dropped = colnames(x)[!keep]
  )
}

apply_scaler <- function(x, prep) {
  x <- as.matrix(x)[, prep$keep, drop = FALSE]
  x <- sweep(x, 2L, prep$center, "-")
  sweep(x, 2L, prep$scale, "/")
}

make_foldid <- function(n, k = 5L, seed) {
  if (k < 3L || k > n) stop("Require 3 <= k <= number of training rows.")
  set.seed(seed)
  sample(rep(seq_len(k), length.out = n))
}

fit_cv_glmnet <- function(x, y, alpha, foldid) {
  glmnet::cv.glmnet(
    x = x,
    y = y,
    family = "gaussian",
    alpha = alpha,
    foldid = foldid,
    standardize = FALSE,
    intercept = TRUE,
    type.measure = "mse"
  )
}

score_regression <- function(y, prediction) {
  y <- as.numeric(y)
  prediction <- as.numeric(prediction)
  if (length(y) != length(prediction) || any(!is.finite(c(y, prediction)))) {
    stop("Metrics require equal-length finite vectors.")
  }
  c(RMSE = sqrt(mean((y - prediction)^2)),
    MAE = mean(abs(y - prediction)))
```

```
}

count_nonzero <- function(fit, s = "lambda.min", tol = 1e-8) {
  b <- as.matrix(stats::coef(fit, s = s))
  slopes <- b[rownames(b) != "(Intercept)", 1L]
  sum(abs(slopes) > tol)
}

safe_condition_numbers <- function(x, lambda, tol = 1e-10) {
  eigenvalues <- eigen(
    crossprod(x) / nrow(x),
    symmetric = TRUE,
    only.values = TRUE
  )$values
  largest <- max(eigenvalues)
  smallest <- max(min(eigenvalues), 0)
  cutoff <- tol * max(1, largest)
  c(
    gram = if (smallest <= cutoff) Inf else largest / smallest,
    ridge = (largest + lambda) / (smallest + lambda)
  )
}
```

1 Problem 1. Prediction design and leakage control

1.1 1A. Target and predictors

```
config <- list(
  file = "prostate.csv",
  response = "lpsa",
  excluded = "lpsa"
)
data_raw <- read.csv(find_exam_data(config$file), check.names = FALSE)
predictors <- setdiff(names(data_raw), config$excluded)
analysis_data <- data_raw[, c(config$response, predictors), drop = FALSE]

if (anyNA(analysis_data)) stop("Declare a training-only missing-value plan.")
if (!all(vapply(analysis_data, is.numeric, logical(1)))) {
  stop("The assigned response and predictors must be numeric.")
}
```

Define the prediction task and justify every exclusion.

1.2 1B. Split and train-only preprocessing

```r
split <- split_rows(nrow(analysis_data), seed = seeds$split)
train_id <- split$train
test_id <- split$test

x_raw <- as.matrix(analysis_data[, predictors, drop = FALSE])
y_raw <- analysis_data[[config$response]]

x_prep <- fit_scaler(x_raw[train_id, , drop = FALSE])
x_train <- apply_scaler(x_raw[train_id, , drop = FALSE], x_prep)
x_test <- apply_scaler(x_raw[test_id, , drop = FALSE], x_prep)
y_train <- y_raw[train_id]
y_test <- y_raw[test_id]

foldid <- make_foldid(nrow(x_train), k = 5L, seed = seeds$cv)
```

Report seeds, dimensions, scaling, folds, and leakage safeguards.

1.3 1C. Training-data exploration

```r
# Use analysis_data[train_id, ] only. Add focused plots and summaries.
```

Interpret multicollinearity, one anomaly, and the question regularization will address.

2 Problem 2. OLS, ridge, and lasso

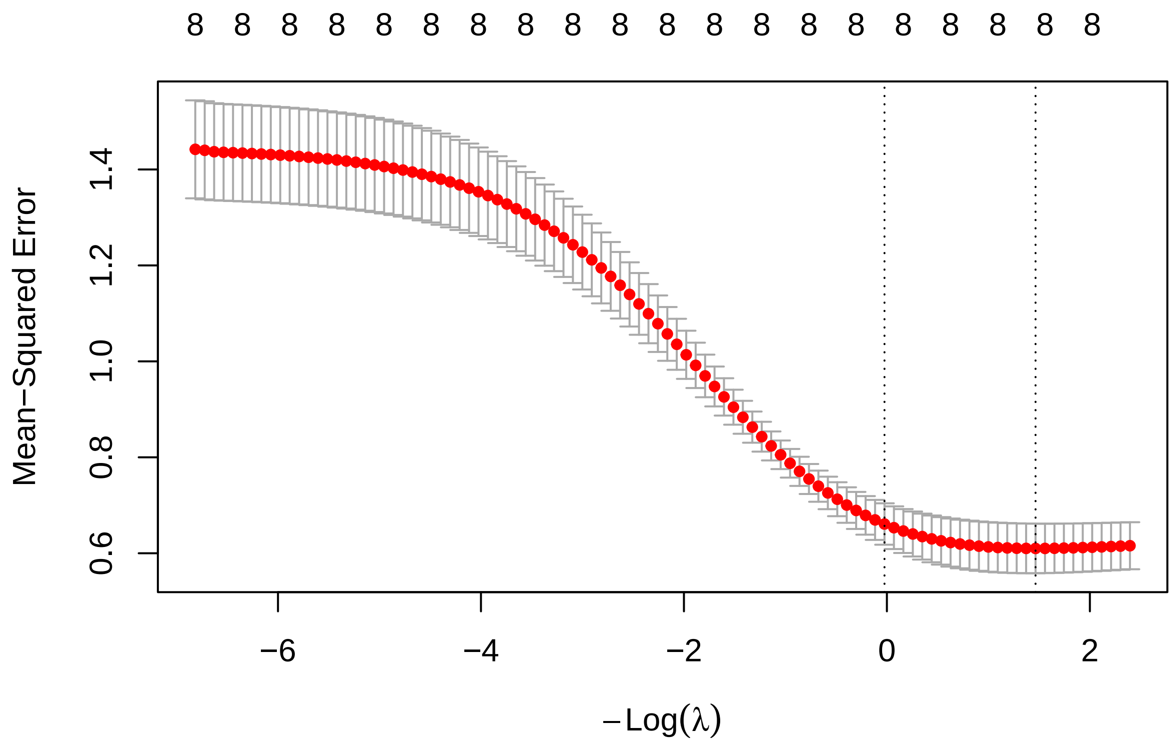
2.1 2A. OLS baseline

```r
# Fit OLS on x_train and y_train. Do not access y_test here.
ols <- lm(y_train ~ x_train)
```

2.2 2B. Ridge

2.2.1 Cross-validation curve:

```r
ridge <- fit_cv_glmnet(x = x_train, y = y_train, alpha=0, foldid = foldid)
plot(ridge)
```



The mean cross-validation MSE decreases as $-Log(\lambda)$ increases (equivalently, as λ decreases), suggesting that reducing the regularization strength improves predictive performance. The left dashed line indicates λ_{1se}, which selects a more regularized model with prediction error within one standard error of the minimum, while the right dashed line indicates λ_{min}, which yields the lowest cross-validation error.

2.2.2 Lambda min:

```
g <- crossprod(x_train)
eigen <- eigen(g)$value

lambda <- ridge$lambda.min
cat("Lambda min: ",lambda, "\n")
```

```
## Lambda min:  0.2311277
```

```
cat("MSE: ", ridge$cvm[ridge$lambda == lambda], "\n")
```

```
## MSE:  0.6100275
```

```r
cat("Condition number:", (max(eigen) + lambda)/(min(eigen) + lambda), "\n")
```

```
## Condition number: 20.22951
```

With $\lambda_{min} \approx 0.23$, the 5-fold cross-validation MSE reaches its minimum value of $MSE_{min} \approx 0.61$, indicating the best predictive performance among the candidate values of λ. However, the corresponding Ridge model still has a relatively large condition number, suggesting that the model remains sensitive to multicollinearity and may not be sufficiently stable.

2.2.3 Lambda 1se:

```r
g <- crossprod(x_train)
eigen <- eigen(g)$value

lambda <- ridge$lambda.1se
cat("Lambda 1se: ",lambda, "\n")
```

```
## Lambda 1se:  1.024039
```

```r
cat("MSE: ", ridge$cvm[ridge$lambda == lambda], "\n")
```

```
## MSE:  0.6609706
```

```r
cat("Condition number:", (max(eigen) + lambda)/(min(eigen) + lambda), "\n")
```

```
## Condition number: 19.06877
```

Although the one-standard-error rule selected a larger penalty $\lambda_{1se} \approx 1.02$, the condition number decreased only slightly from 20.2 to 19.07, indicating minimal improvement in numerical stability. In contrast, the cross-validation MSE increased from 0.61 to 0.66. Therefore, the additional regularization provided by λ_{1se} offers little practical benefit while reducing predictive performance.

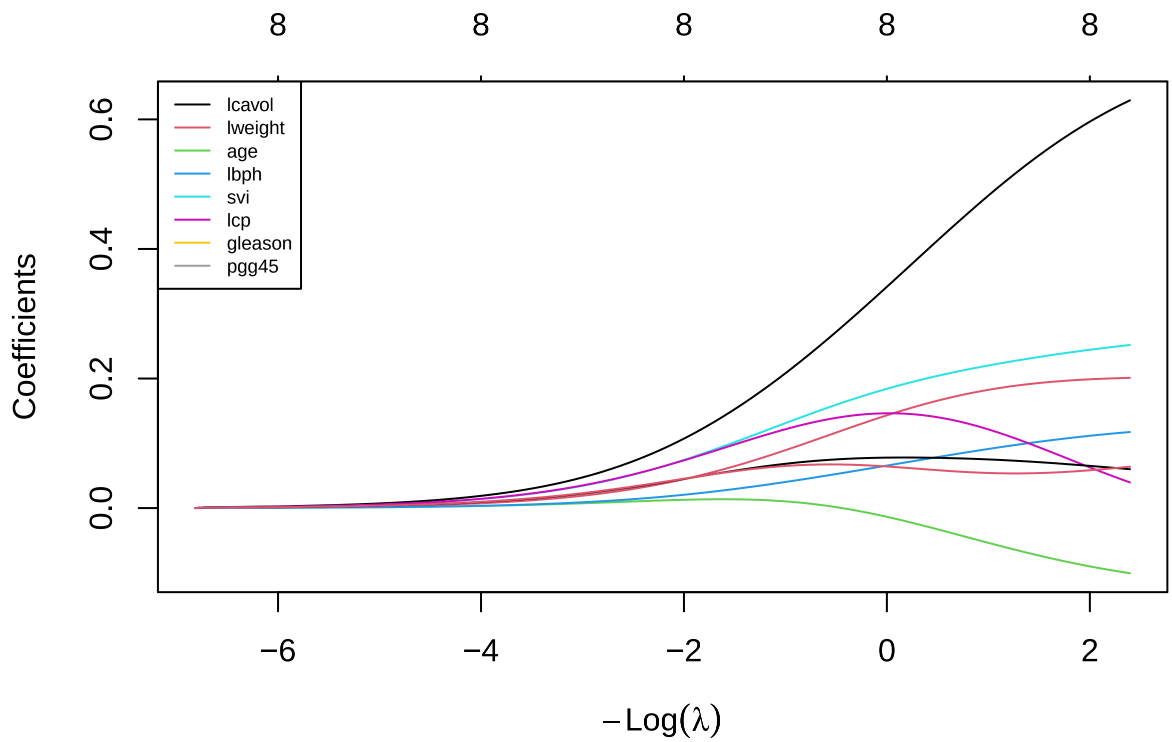
2.2.4 Coefficients:

```r
data.frame(
  OLS = as.numeric(coef(ols)),
  Lambda_Min = as.numeric(coef(ridge, "lambda.min")),
  Lambda_Max = as.numeric(coef(ridge, "lambda.1se")),
  row.names = names(ols$coefficients)
)
```

```
##                       OLS  Lambda_Min  Lambda_Max
## (Intercept)     2.47569468  2.47569468  2.47569468
## x_trainlcavol   0.72282252  0.54055005  0.33822482
## x_trainlweight  0.20222141  0.19253291  0.14197627
## x_trainage     -0.13046518 -0.07201800 -0.01262288
## x_trainlbph     0.13361911  0.10166159  0.06498686
## x_trainsvi      0.27494937  0.23239051  0.18307566
## x_trainlcp     -0.04125506  0.09593321  0.14631601
## x_traingleason  0.03648450  0.07123914  0.07784505
## x_trainpgg45    0.09450125  0.05388195  0.06447679
```

Compared with the OLS model, Ridge regression shrinks most regression coefficients toward zero, reducing their magnitudes through the regularization penalty. This shrinkage helps alleviate the effects of multicollinearity while retaining all predictors in the model. Most coefficients exhibit the expected behavior by moving closer to zero as the regularization strength increases. However, two coefficients show notable exceptions. The coefficient of **lcp** changes from a small negative value in the OLS model to positive values under Ridge regression. In addition, the coefficient of **gleason** increases in magnitude, moving further away from zero rather than shrinking like the other coefficients. These changes suggest that the effects of **lcp** and **gleason** are influenced by multicollinearity with other predictors. As Ridge simultaneously estimates all coefficients under a penalty, it redistributes the effects among correlated variables, causing some coefficients to increase in magnitude or even change sign despite the overall shrinkage trend.

```r
n_vars <- nrow(ridge$glmnet.fit$beta)
plot(ridge$glmnet.fit)
legend("topleft",
       legend = rownames(ridge$glmnet.fit$beta),
       col = 1:n_vars,
       lty = 1,
       cex = 0.6)
```



As λ increases, all regression coefficients are progressively shrunk toward zero, demonstrating the regularization effect of Ridge regression. Conversely, as λ decreases, the coefficients gradually move away from zero toward their OLS estimates. The coefficient of **lcavol** remains the largest throughout the regularization path, indicating that it has the strongest association with the response variable. The coefficients of **svi** and lweight also increase substantially as the regularization weakens, suggesting relatively strong effects on the response. In contrast, the coefficients of **gleason**, **lbph**, and **pgg45** remain relatively small across the entire path, indicating weaker contributions to the model. The coefficient of age remains negative throughout the regularization path, reflecting a consistent negative relationship with the response variable. Notably, the coefficient of **lcp** exhibits non-monotonic behavior, first increasing and then decreasing as λ changes, suggesting that its estimated effect is sensitive to multicollinearity with other predictors.

To address multicollinearity, Ridge impose an L_2 penalty on the regression coefficients. Rather than assigning a large coefficient to one correlated predictor and a small coefficient to another, Ridge shrinks the coefficients simultaneously and redistributes their effects among the correlated variables. Consequently, the coefficient estimates become more stable and less sensitive to small changes in the data.

2.3 2C. Lasso

```
# Use fit_cv_glmnet(..., alpha = 1, foldid = foldid).
# Report lambda.min, lambda.1se, paths, and nonzero predictors.
```

2.4 2D. Training-based comparison and locked core model

```
# Build a comparison from training/CV quantities only.
```

Core model nominated before holdout evaluation: [method and reason]

2.5 2E. Perturbation or stability check

Prediction before running the check: [prediction]

```
# Implement one allowed training-only stability check.
```

Observed result and explanation: [result]

3 Problem 3. Mathematical mechanisms

Let

$$X \in \mathbb{R}^{n \times p}$$

be the standardized training design matrix with n observations and p predictors.
Let

$$y \in \mathbb{R}^{n}$$

be the response vector and

$$b \in \mathbb{R}^{p}$$

be the coefficient vector.

3.1 3A. Ridge and conditioning

Write the ridge derivation. Define dimensions and explain uniqueness.

```
# Use x_train and the fitted ridge lambda.min.
# safe_condition_numbers(x_train, lambda = ...)
```

3.2 3A. Ridge and conditioning

3.2.1 Ridge objective

The ridge objective function is

$$Q_\lambda(b) = \frac{1}{2}\|y - Xb\|^2 + \frac{\lambda}{2}\|b\|^2.$$

Taking the gradient,

$$\nabla Q_\lambda(b) = (X^T Xb - X^T y) + \lambda b.$$

Setting the gradient equal to zero,

$$(X^T X + \lambda I)b = X^T y.$$

Define

$$G = X^T X, \qquad z = X^T y.$$

Hence,

$$(G + \lambda I)b = z,$$

and therefore

$$\boxed{\hat{b} = (G + \lambda I)^{-1} z.}$$

Since G is positive semidefinite and $\lambda > 0$,

$$G + \lambda I$$

is positive definite and therefore invertible. Hence the ridge estimator is unique.

The spectral condition number is

$$\kappa_2(G) = \frac{\mu_{\max}}{\mu_{\min}},$$

while ridge gives

$$\kappa_2(G + \lambda I) = \frac{\mu_{\max} + \lambda}{\mu_{\min} + \lambda}.$$

where

$$\mu_{\max} = \max_i \mu_i$$

is the **largest eigenvalue** of the Gram matrix G, and

$$\mu_{\min} = \min_i \mu_i$$

is the **smallest (positive) eigenvalue** of G.

3.2.2 Condition Number and Numerical Stability

The Gram matrix is defined as

$$G = \frac{1}{n} X^T X.$$

Since G is symmetric and positive semidefinite, all of its eigenvalues are non-negative. Let the eigenvalues of G be

$$\mu_1, \mu_2, \cdots, \mu_p,$$

where

$$\mu_{\max} = \max_i \mu_i, \qquad \mu_{\min} = \min_i \mu_i.$$

The spectral condition number is defined as

$$\kappa_2(G) = \frac{\mu_{\max}}{\mu_{\min}}.$$

This quantity measures how sensitive the solution of a linear system is to small perturbations in the input data. A large condition number indicates that the Gram matrix is close to singular, so the estimated coefficients may change dramatically even when the training data are only slightly perturbed. Conversely, a small condition number indicates that the matrix is well-conditioned and the regression coefficients are numerically stable.

Ridge regression replaces the Gram matrix by

$$G + \lambda I.$$

If the eigenvalues of G are

$$\mu_1, \mu_2, \cdots, \mu_p,$$

then the eigenvalues of the ridge matrix become

$$\mu_1 + \lambda,\ \mu_2 + \lambda,\ \dots,\ \mu_p + \lambda.$$

Hence, the condition number becomes

$$\kappa_2(G + \lambda I) = \frac{\mu_{\max} + \lambda}{\mu_{\min} + \lambda}.$$

Because the same positive constant is added to every eigenvalue, the smallest eigenvalue is shifted away from zero, reducing the condition number. Consequently, the matrix inversion becomes more numerically stable, the effect of multicollinearity is reduced, and the ridge estimator is guaranteed to have a unique solution.

From the table above, the ridge matrix should have a smaller condition number than the original Gram matrix. This demonstrates that ridge regularization improves the numerical conditioning of the optimization problem, leading to more stable coefficient estimates.

3.3 3B. Lasso optimality and soft thresholding

Derive the zero/nonzero optimality conditions and the orthonormal-design formula.

3.4 3B. Lasso optimality and soft thresholding

Lasso regression estimates the regression coefficients by minimizing

$$Q_\lambda(b) = \|y - Xb\|^2 + \lambda\|b\|,$$

where

$$\|b\| = \sum_{j=1}^{p} |b_j|,$$

and $\lambda > 0$ controls the amount of regularization.

Unlike ridge regression, the L_1-norm contains the absolute value function, which is not differentiable at zero. Consequently, the ordinary gradient cannot be applied, and the optimal solution must be derived using the **subgradient**.

3.4.1 Step 1. Rewrite the Objective Function

Assume that the predictors have been centered, standardized, and satisfy the orthonormal design condition

$$X^T X = I.$$

Define

$$z = X^T y.$$

The squared error term can be expanded as

$$\begin{aligned}\|y - Xb\|^2 &= (y - Xb)^T (y - Xb) \\ &= y^T y - 2b^T X^T y + b^T X^T X b.\end{aligned}$$

Using

$$X^T X = I,$$

the objective becomes

$$Q_\lambda(b) = \frac{1}{2} b^T b - b^T z + \lambda \sum_{j=1}^{p} |b_j| + C,$$

where

$$C = y^T y$$

is a constant independent of b.

Expanding by coordinates,

$$Q_\lambda(b) = \sum_{j=1}^{p} \left[\frac{1}{2} b_j^2 - z_j b_j + \lambda |b_j| \right] + C.$$

Therefore, each coefficient can be optimized independently.

3.4.2 Step 2. Compute the Subgradient

Since

$$|b_j|$$

is not differentiable at zero, its subgradient is

$$\partial |b_j| = \begin{cases} 1, & b_j > 0, \\ [-1, 1], & b_j = 0, \\ -1, & b_j < 0. \end{cases}$$

The first-order optimality condition becomes

$$0 \in b_j - z_j + \lambda \partial |b_j|.$$

The solution is obtained by considering three cases.

3.4.2.1 Case 1. Positive coefficient

Suppose

$$b_j > 0.$$

Then

$$\partial|b_j| = 1,$$

and

$$b_j - z_j + \lambda = 0.$$

Hence

$$b_j = z_j - \lambda.$$

For this solution to remain positive,

$$z_j > \lambda.$$

3.4.2.2 Case 2. Negative coefficient

Suppose

$$b_j < 0.$$

Then

$$\partial|b_j| = -1,$$

and

$$b_j - z_j - \lambda = 0.$$

Therefore,

$$b_j = z_j + \lambda.$$

Since the coefficient must remain negative,

$$z_j < -\lambda.$$

3.4.2.3 Case 3. Zero coefficient

Suppose

$$b_j = 0.$$

The subgradient condition becomes

$$0 \in -z_j + \lambda[-1, 1].$$

This is satisfied whenever

$$|z_j| \leq \lambda.$$

Therefore,

$$b_j = 0.$$

This explains why lasso can produce exact zero coefficients.

3.4.3 Step 3. Soft-Thresholding Solution

Combining the three cases,

$$\hat{b}_j = \begin{cases} z_j - \lambda, & z_j > \lambda, \\ 0, & |z_j| \leq \lambda, \\ z_j + \lambda, & z_j < -\lambda. \end{cases}$$

The piecewise solution can be written compactly as

$$\boxed{\hat{b}_j = \operatorname{sign}(z_j)\left(|z_j| - \lambda\right)_+}$$

where

$$(a)_+ = \max(a, 0).$$

This expression is known as the **soft-thresholding operator**.

3.4.4 Interpretation

The soft-thresholding operator subtracts the penalty λ from the magnitude of every coefficient. When

$$|z_j| \le \lambda,$$

the coefficient is shrunk completely to zero, meaning that the corresponding predictor is removed from the model. Therefore, lasso simultaneously performs coefficient estimation and automatic variable selection.

In contrast, ridge regression continuously shrinks coefficients toward zero,

$$\hat{b}_{\text{ridge}} = (G + \lambda I)^{-1} z,$$

but almost never makes them exactly zero. Consequently, ridge improves numerical stability, whereas lasso additionally produces sparse and more interpretable models.

3.5 3C. Connect algebra to evidence

Connect one specific numerical result to Problem 3A or 3B.

4 Problem 4. Elastic net and a neural-feature bridge

4.1 4A. Research question and source map

Research direction: [weight decay / sparse weights / pruning / dropout]

Claim	Exact primary source and locator	What it establishes	What it does not establish
[Claim]	[Citation, section/page/equation]	[Support]	[Boundary]

4.2 4B. Elastic net on original predictors

```r
alpha_grid <- c(0.10, 0.25, 0.50, 0.75, 0.90)

# Fit one cv.glmnet object per alpha with the same foldid.
# Select alpha and lambda using training CV only.
```

4.3 4C. Fixed ReLU features

```r
relu <- function(z) pmax(z, 0)
M <- 200L

# TODO:
# 1. Generate A and bias once from seeds$features using the stated distributions.
# 2. Apply the same A and bias to x_train and x_test.
# 3. Fit a scaler to H_train only and apply it to H_test.
# 4. Fit ridge, lasso, and elastic net on H_train with the shared foldid.
```

State which weights are fixed, which are learned, and how active features are counted.

4.4 4D. Locked declaration and final holdout evaluation

> **Locked before viewing `y_test`:** [preprocessing, candidate specifications, selected tuning values, feature seed, nominated deployment model, metrics]

```r
# This must be the first analysis chunk that uses y_test for evaluation.
# Generate all fixed candidate predictions and one RMSE/MAE table here.
```

State whether the holdout evidence supports the predeclared choice. Do not retune.

4.5 4E. Deep-learning connection and limitations

Distinguish penalty, weight decay, output-weight sparsity, pruning, dropout, and a trained deep network. Cite the sources from 4A and discuss at least two limitations.

5 Problem 5. Report quality and reproducibility

5.1 5A. Reproduction record

```
1. Open a clean R session.
2. [Exact commands and folder layout.]
3. Render GroupXX_ProjectQuiz1.Rmd.
```

5.2 5B. Recommendation and limitations

Give the final recommendation, distinguish prediction from interpretation, and state limitations.

5.3 5C. AI verification record

AI-assisted item	How it was checked	Evidence	Modification or rejection
[Item]	[Method]	[Test/source/output]	[Decision]

Preflight and session information

```r
# Add every object required to reproduce the final table and figures.
required_objects <- c(
  "train_id", "test_id", "x_train", "x_test", "y_train", "y_test", "foldid"
)
missing_objects <- setdiff(required_objects, ls())
if (length(missing_objects)) {
  stop(
    "Missing required objects: ",
    paste(missing_objects, collapse = ", ")
  )
}
stopifnot(length(intersect(train_id, test_id)) == 0L)
```

```r
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 26.04 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.32.so;  LAPACK version
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
##  [3] LC_TIME=en_US.UTF-8        LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=en_US.UTF-8    LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=en_US.UTF-8       LC_NAME=C
##  [9] LC_ADDRESS=C               LC_TELEPHONE=C
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Asia/Ho_Chi_Minh
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
```

```
##  [1] knitr_1.51      broom_1.0.13    glmnet_5.0      Matrix_1.7-4
##  [5] lubridate_1.9.5 forcats_1.0.1   stringr_1.6.0   dplyr_1.2.1
##  [9] purrr_1.2.2     readr_2.2.0     tidyr_1.3.2     tibble_3.3.1
## [13] ggplot2_4.0.3   tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] generics_0.1.4     shape_1.4.6.1      stringi_1.8.7      lattice_0.22-9
##  [5] hms_1.1.4          digest_0.6.39      magrittr_2.0.5     timechange_0.4.0
##  [9] evaluate_1.0.5     grid_4.5.2         RColorBrewer_1.1-3 iterators_1.0.14
## [13] fastmap_1.2.0      foreach_1.5.2      backports_1.5.1    survival_3.8-6
## [17] scales_1.4.0       codetools_0.2-20   cli_3.6.6          rlang_1.2.0
## [21] splines_4.5.2      withr_3.0.2        yaml_2.3.12        otel_0.2.0
## [25] tools_4.5.2        tzdb_0.5.0         vctrs_0.7.3        R6_2.6.1
## [29] lifecycle_1.0.5    pkgconfig_2.0.3    pillar_1.11.1      gtable_0.3.6
## [33] glue_1.8.1         Rcpp_1.1.2         xfun_0.58          tidyselect_1.2.1
## [37] rstudioapi_0.18.0  farver_2.1.2       htmltools_0.5.9    rmarkdown_2.31
## [41] compiler_4.5.2     S7_0.2.2
```